

Options	Description
-o	Write information on outstanding usage. (Number of messages on queue and total number of bytes in messages on queue for message queues, and number of processes attached to shared memory segments.)
-p	Write process number information. (Process ID of the last process to send a message and process ID of the last process to receive a message on message queues, process ID of the creating process, and process ID of the last process to attach or detach on shared memory segments.)
-t	Write time information. (Time of the last control operation that changed the access permissions for all facilities, time of the last msgsnd() and msgrcv() operations on message queues, time of the last shmat() and shmdt() operations on shared memory, and time of the last semop() operation on semaphores.)

(VII) Recourses Required

- Computer System with basic configuration. Linux/Ubuntu/ CentOS /any other open source Operating System.

(VIII) Precautions to be followed

- (1) Handle computer system with care.
- (2) Strictly follow the instructions for writing commands
- (3) Follow safety Practices

(IX) Conclusion

Successfully executed a) Commands for sending messages to logged in use - who, cat, wall, write, msg.
b) List processes attached to a shared memory

(X) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) What are the prerequisites for using write command?
- (2) Write a command to display the current run level using who command?
- (3) Write a command to append the contents of one file to another using cat Command?
- (4) Write a command to send a message from a file using wall command?
- (5) Write a command to check the current message permission status of your terminal?
- (6) What is the purpose of the ipcs command in Linux?

[Space to Write for Answers]

1) Both users must be logged into the system.
The receiver must allow messages (msg y).

2) who -r

3) cat file1 >> file2

4) wall < message.txt

5) msg

6) The `ipcs` command displays information about Inter-Process Communication (IPC) resources such as shared memory segments, message queues and semaphore stats.